

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

a) Write a C code to create a process using fork() system call.

b) Read the user input as shell command.

c) Use a system call exec() to execute the shell command.

e) Display the PIDs of parent and child processes.

```
Last login: Wed Aug  5 08:31:05 on ttys000
vedadripinninti@vedadris-MacBook-Air ~ % nano PIDs.c
vedadripinninti@vedadris-MacBook-Air ~ %
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 100
#define MAX_ARGS 20

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);

    // Remove newline character
    input[strcspn(input, "\n")] = '\0';

    // Split the command into arguments
    int i = 0;
    args[i] = strtok(input, " ");

    while (args[i] != NULL && i < MAX_ARGS - 1) {
        i++;
        args[i] = strtok(NULL, " ");
    }

    pid_t pid = fork();

    if (pid < 0) {
        // Fork failed
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
```

1)Then press ctrl+oenter ctrl+x2)then gcc filename.c -o filename

```
Last login: Wed Aug  5 08:31:05 on ttys000
vedadripinninti@vedadris-MacBook-Air ~ % nano PIDs.c
vedadripinninti@vedadris-MacBook-Air ~ %
```

3)then ./filename

```
Last login: Wed Aug  5 08:31:05 on ttys000
vedadripinninti@vedadris-MacBook-Air ~ % nano PIDs.c
vedadripinninti@vedadris-MacBook-Air ~ % ./PIDs
Enter a Linux command:
```

4)enter the linux command

```
Enter a Linux command: ls

Parent Process
Parent PID : 10314
Child PID  : 10315

Child Process
Child PID  : 10315
Parent PID : 10314
<!doctype html>.html          file.c          Passbook.jpg
2ebc9504-640c-4e44-9851-d67271161b01.JPG  Home page.html Pictures
airline                        IMG_8048.jpg    PiDs
Applications                   IMG_8049.heic   PIDs.c
Desktop                        IMG_8050.JPG    process
Documents                      Library         process.c
Downloads                      Movies          project1.1
file                           Music           Public
Child process completed.
vedadripinninti@vedadris-MacBook-Air ~ %
```

Child process completed.

Task 2

Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

An operating system is system software that controls all the hardware and software resources of a computer. Users do not communicate directly with the hardware. Instead, the operating system acts as a bridge between

the user, applications, and the hardware
Linux Commands Used

1. uname

The `uname` command is used to display information about the Linux operating system. It provides details such as the operating system name, kernel version, machine architecture, and hostname. This command helps us identify the system on which Linux is running.

```
vedadripinninti@vedadris-MacBook-Air ~ % uname
Darwin
vedadripinninti@vedadris-MacBook-Air ~ %
```

2. lscpu

The `lscpu` command displays detailed information about the processor. It shows the CPU model, number of cores, number of threads, CPU architecture, and processor speed. This information helps us understand the hardware capabilities of the computer.

```
vedadripinninti@vedadris-MacBook-Air ~ % uname
Linux
vedadripinninti@vedadris-MacBook-Air ~ % lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:              39 bits physical, 48 bits virtual
  Byte Order:                 Little EndianCPU(s):                12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
Model name:                  12th Gen Intel(R) Core(TM) i5-1235U
CPU family:                  6
Model:                       154
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                   1
Stepping:                    4
BogoMIPS:                    4992.00
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca
cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm
constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq
pni pclmulqdq vmx sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt
tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch
ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust
bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt
xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri
movdir64b fsrm md_clear serialize ibt flush_lld arch_capabilities
Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
```

3. ps

The ps command lists the processes that are currently running in the system. It displays the Process ID (PID), Parent Process ID (PPID), user name, and the command associated with each process. This command is useful for monitoring and managing processes.

```
vedadripinninti@vedadris-MacBook-Air ~ % ps
  PID TTY          TIME CMD
  426 pts/0        00:00:00 bash
  476 pts/0        00:00:00 ps
vedadripinninti@vedadris-MacBook-Air ~ %
```

4. top

The top command provides a real-time view of the system's performance. It continuously displays CPU usage, memory usage, running processes, and system load. It is commonly used to identify programs that consume a large amount of system resources.

```
top - 07:13:17 up 29 min, 1 user, load average: 0.00, 0.02, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3757.6 total, 3188.2 free, 396.2 used, 247.8 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3361.5 avail Mem

  PID USER   PR NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
    1 root    20  0   2184 13080 9764 S   0.0  0.3   0:00.51 systemd
    2 root    20  0    3180 2208 2072 S   0.0  0.1   0:00.01 init-systemd(Ub
    7 root    20  0    3196 2128 2008 S   0.0  0.1   0:00.00 init
   54 root    19 -1 50312 16212 15148 S   0.0  0.4   0:00.14 systemd-journal
  100 root    20  0   25052 6540 5224 S   0.0  0.2   0:00.16 systemd-udev
  116 systemd+ 20  0   21344 13292 11164 S   0.0  0.3   0:00.08 systemd-resolve
  117 systemd+ 20  0   91036 8076 7104 S   0.0  0.2   0:00.10 systemd-timesyn
  166 root    20  0    4244 2752 2496 S   0.0  0.1   0:00.00 cron
  167 message+ 20  0    9636 5452 4756 S   0.0  0.1   0:00.05 dbus-daemon
  174 root    20  0   17976 8864 7820 S   0.0  0.2   0:00.05 systemd-logind
  176 root    20  0 1830616 14320 11296 S   0.0  0.4   0:00.07 wsl-pro-service
  181 syslog   20  0   222516 5688 4680 S   0.0  0.1   0:00.04 rsyslogd
  193 root    20  0    3124 2040 1900 S   0.0  0.1   0:00.00 agetty
  199 root    20  0 107036 23416 13964 S   0.0  0.6   0:00.10 unattended-upgr
  294 root    20  0    6704 4740 3952 S   0.0  0.1   0:00.00 login
  342 vedad+  20  0   20332 11652 9520 S   0.0  0.3   0:00.12 systemd
  343 vedad+  20  0   21168 3572 1848 S   0.0  0.1   0:00.00 (sd-pam)
  367 vedad+  20  0    6080 5368 3700 S   0.0  0.1   0:00.01 bash
  418 root    20  0    3188 1120 980 S   0.0  0.0   0:00.00 SessionLeader
  420 root    20  0    3204 1260 1108 S   0.0  0.0   0:00.15 Relay(426)
  426 vedad+  20  0    6080 5420 3704 S   0.0  0.1   0:00.07 bash
  479 vedad+  20  0    9324 5776 3568 R   0.0  0.2   0:00.07 top
```

